MILITARY SPECIFICATION
FIN, RETARDER, AIR INFLATABLE
BSU-85/B

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the performance, design, development, and test requirements for a low-drag stabilizer and high-drag retarder (BSU-85/B) for attachment to the MK 83 bomb.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified the following specifications, standards and handbooks of the issue listed in that issue of the Department of the Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-A-8591	Airborne Stores and Associated Suspension Equipment, General Design Criteria for
MIL-W-8611	Welding, Metal, Arc and Gas, Steel and Corrosion and Heat Resistent Alloys, Process for
MIL-P-9024	Packaging, Handling and Transportability in System/Equipment Acquisition
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities

Beneficial Comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

2.1.1 Specifications, Standards, and Handbooks - continued

STANDARDS

Federal

FED-STD-102	Preservation, Packaging and Packing Levels

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-794	Parts and Equipment, Procedures for Packaging of
MIL-STD-810	Environmental Test Methods
MIL-STD-882	System Safety Program Requirements
MIL-STD-889	Dissimilar Metals
MIL-STD-965	Parts Control Program
MIL-STD-1289	Ground Fit and Compatibility Tests of Airborne Stores, Procedure for
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment Facilities
MIL-STD-1523	Age Controls of Age-Sensitive Elastomeric Material
MIL-STD-1763	Aircraft/Stores Certification Procedures
MIL-STD-45662	Calibration System Requirements

HANDBOOKS

Military

MIL-HDBK-5	Metallic Materials and Elements for Aerospace Vehicle Structures
MIL-HDBK-17	Plastics for Aerospace Vehicles Notice #1, #11 Part 1 Reinforced Plastics
MIL-HDBK-23	Structural Sandwich Composites

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings and publications. The following drawings form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command
(Code Indent 30003)

2841464, 982537	Bomb, General Purpose, 1000 Pound, MK 83, Mod 4, Loaded
2841464, 4902089	Bomb, General Purpose, 1000 Pound, MK 83, Mod 5, Thermally-Coated, Loaded

(Copies of specifications, standards, handbooks, drawings, and publications, required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Item description. The BSU-85/B stabilizer/retarder, hereinafter referred to as "the item", shall be an attachment to the MK 83 GP Bomb. The item shall consist of a low-drag stabilizer assembly, a deployable retarder, and a lanyard operated deployment system. The combination of the item with either configuration of the MK 83 bomb listed in 3.1.2.1 shall make up the system.

3.1.1 Item diagrams. The air carriage release envelope diagram is shown in FIGURE 1.

3.1.2 Interface description.

3.1.2.1 Item to MK 83 bomb. The item shall be capable of attaching to the MK 83 bomb body in accordance with Drawing 982537 and Drawing 4902089 by interface with the attaching ring. The item shall incorporate an indexing provision for the purpose of orienting the fins at 45 degrees relative to the bomb's suspension lugs.

3.1.2.2 Item to launch aircraft. The item, when attached to the MK 83 bomb, shall be capable of being loaded and carried on the Multiple Ejector Racks (MERs), Improved Multiple Ejector Racks (IMER), Triple Ejection Racks (TERs), Improved Triple Ejector Racks (ITER) Vertical Ejector Rack (VER) and 1000 pound compatible parent racks on the A-4, A-6, A-7, AV-8, F-4, P-3, S-3, and F/A 18 aircraft. The bomb rack compatibility requirements shall be as specified in TABLE I. Procedures for ground fit and compatibility tests shall be in accordance with MIL-STD-1289. Deployment of the retarder shall be controlled by a lanyard. The lanyard shall attach by means of a frangible link to an arming solenoid on the bomb rack. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

TABLE I. MK 83/BSU-85/B air bomb rack/store compatibility requirements.

AIRCRAFT	SERVICE	BOMB RACKS
A-4	(N) (M)	AERO-7A, AERO-20A, MER 7, TER-7, IMER, ITER
A-6	(N) (M)	AERO-7A, AERO-7B, MER-7, TER-7 IMER, ITER
A-7	(N)	BRU-10, MER-7 TER-7, IMER, ITER
AV-8 A/C	(M)	ERU-119
AV-8B	(M)	BRU-36, TER-7, ITER
F-4	(M)	MER-7, TER-7, IMER, ITER
F/A-18	(N)(M)	BRU-32, TER-7, BRU-33 (VER), ITER
P-3	(N)	BRU-15 External Carriage Only
S-3	(N)	BRU-11, TER-7 ITER External Carriage Only

(N) = Navy
(M) = Marines

frangible link shall have a solenoid attachment loop of .078 plus or minus .002 inch diameter material, with a radius of .375 minus .010 inch. When the system is released with the arming solenoid unenergized, the retarder shall not deploy, and the system shall separate from the aircraft in the low drag mode. When released with the arming solenoid energized, the system shall separate from the aircraft in the high drag mode. No part of the lanyard other than the attachment link shall remain with the aircraft after release. The interface lanyard pull force shall be between 25 and 150 pounds and operate with existing bomb rack solenoid. No debris shall be released by the item during separation of the system from the aircraft in either the low or high drag configuration.

3.1.2.3 Accessory physical interface. The FMU-139, and MK 376 bomb fuzes shall require the use of external arming wires which shall be routed along the bomb and shall pass through the item structure. The item shall have both an entrance and an exit hole for the arming wire. The centerline of these holes shall be located not greater than one inch off a vertical plane through the lug line, and shall be 2.00 plus or minus .25 inches from the bomb/item interface. The holes shall be not less than 0.35 inches in diameter and shall have guides to prevent sharp bending of the arming wire. Nose mechanical fuzing (M904), when used with retarded deliveries, requires that its initiation be dependent on retarder deployment, which shall be accomplished by the use of one arming wire between the nose fuze and the retarder. The point at which this wire is attached to the retarder shall be capable of providing a force of not less than 100 pounds over a stroke of not less than 4.5 inches. The item shall permit installation and routing of arming wires/lanyards after the item has been attached to the bomb.

3.1.2.4 Fuze installation. Access to those items installed in the rear fuze well, MK 376 and FMU-139 shall be available through the side of the item, so as to permit settings to be made or safety pins removed prior to flight. The access shall be located so that a visual check can be made of the arm/safe status of the fuze when installed in the bomb, and the proper assembly and routing of the arming wire. The access shall allow installation of the fuze in the MK 83 aft fuze well when the item is attached.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

3.1.2.5 Fuze function. The item, when attached to the MK 83 bomb and released at 450 Knots Calibrated Airspeed (KCAS) and above, shall provide deceleration along the bomb longitudinal axis of 4 g's in not greater than 0.57 seconds after release of cover and remain at 4 g's or more for not less than 0.75 seconds. When released at 550 KCAS or less, the deceleration along the bomb longitudinal axis shall be not greater than 13 g's after 0.71 seconds from aircraft release. When the system is released in the low-drag mode within the delivery speed and altitude envelope specified in FIGURE 1, the item shall not produce deceleration along the bomb's longitudinal axis greater than 1.4 g's. The item, when the system is released in the low drag mode, shall not pull the arming wire/lanyard for any retarder initiated fuze.

3.1.3 Major component list. Not applicable.

3.1.4 Government furnished property list. Not applicable.

3.1.5 Government loaned property list. Not applicable.

3.2 Characteristics.

3.2.1 Performance. The item, when attached to the MK-83 bomb, shall perform as specified herein and under the following conditions.

a. Airspeed
   (1) Carriage and release; limits of the basic aircraft (LBA) or 650 KCAS whichever is less.

b. Acceleration
   (1) Carriage - LBA or -3.0 g to +7.5 g whichever is less
   (2) Release - +0.5 g to +5.0 g

c. Maximum dive angle for release:
   (1) Unretarded - 60 degrees
   (2) Retarded - 45 degrees

d. Aircraft Loading
   (1) All 1,000 lb compatible parent racks - 1
   (2) Triple Ejector Rack (TER) - 3
   (3) Multiple Ejector Rack (MER) - 3

e. Minimum Release Interval from the same MER/TER:
   (1) 100 ms desired
   (2) 200 ms required

f. Accuracy
   (1) Unretarded - 4.0 MILS CEP (300-550 KCAS); 5 MILS CEP (550-650 KCAS)
   (2) Retarded - 12.0 MILS CEP (300-550 KCAS); 14 MILS CEP (550-650 KCAS)

g. Fuze Compatibility
   (1) FMU-139 and M904 required
   (2) MK 376 desired

h. High/Low Drag Pilot Option

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

j. Physical/visual access to tail fuze (ensure correct fuze and arming wire installation)

k. Retarder arming wire for arming M904 fuze

m. No debris and all arming wire integral to fin

n. Compatible with existing GP bomb GSE (ground support equipment)

p. Non-explosive AIR system

q. No special tools required

r. Service life of one year

s. Shelf life of ten years

3.2.1.1 Flight. The item shall be capable of meeting all the requirements of 3.2.1 when carried and released within the flight envelope shown in FIGURE 1 and at a maximum aircraft dive angle release of 60 degrees unretarded and 45 degrees retarded.

3.2.1.2 Temperature. The item shall be capable of operation at the high temperature extremes specified in MIL-STD-210, and over the range of dynamic pressures defined by the envelope shown in FIGURE 1.

3.2.1.3 Aircraft separation. The item, when attached to the MK 83, shall separate from the delivery aircraft under the conditions specified in FIGURE 1 such that no contact between the bomb and the aircraft shall occur and no contact shall occur between the bomb and any part of the bomb rack or adjacent stores.

3.2.1.4 Retarder deployment. When released in the high drag mode and within the high-drag envelope of FIGURE 1, the retarder shall deploy in no greater than 1.0 second at 250 KCAS release airspeed. The deployment times at 500 and 650 KCAS release airspeed shall be not greater than 0.500 and 0.400 seconds, respectively. Deployment time is described as the time from the initiation of the retarder deployment process until attainment of maximum drag. The retarder shall remain fully deployed throughout the systems trajectory.

3.2.1.5 Accuracy. The accuracy threshold shall be determined using the procedures and conditions as specified in 4.2.3 and 4.2.4. The accuracy threshold of the system when released in the low-drag mode shall be a ballistic dispersion not greater than 4 mils, Circular Error Probability (CEP) at 300-550 KCAS and not greater than 5 mils CEP at 550-650 KCAS. The accurracy threshold of the system when released in the high-drag mode shall be a ballistic dispersion of not greater than 12 mils CEP at 300-550 KCAS and not greater than 14 mils CEP at 550-650 KCAS.

3.2.1.6 Structural adequacy. The item shall be capable of meeting the load requirements of MIL-A-8591. For the carriage phase, the safety factor between the limit load and the yield load shall be not less than 1.15 and the safety factor between the limit load and the ultimate load shall not be less than 1.5. The item shall be capable of supporting the airloads when the system is released from the aircraft. A factor of safety of 1.0 shall be applied between limit load and yield

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

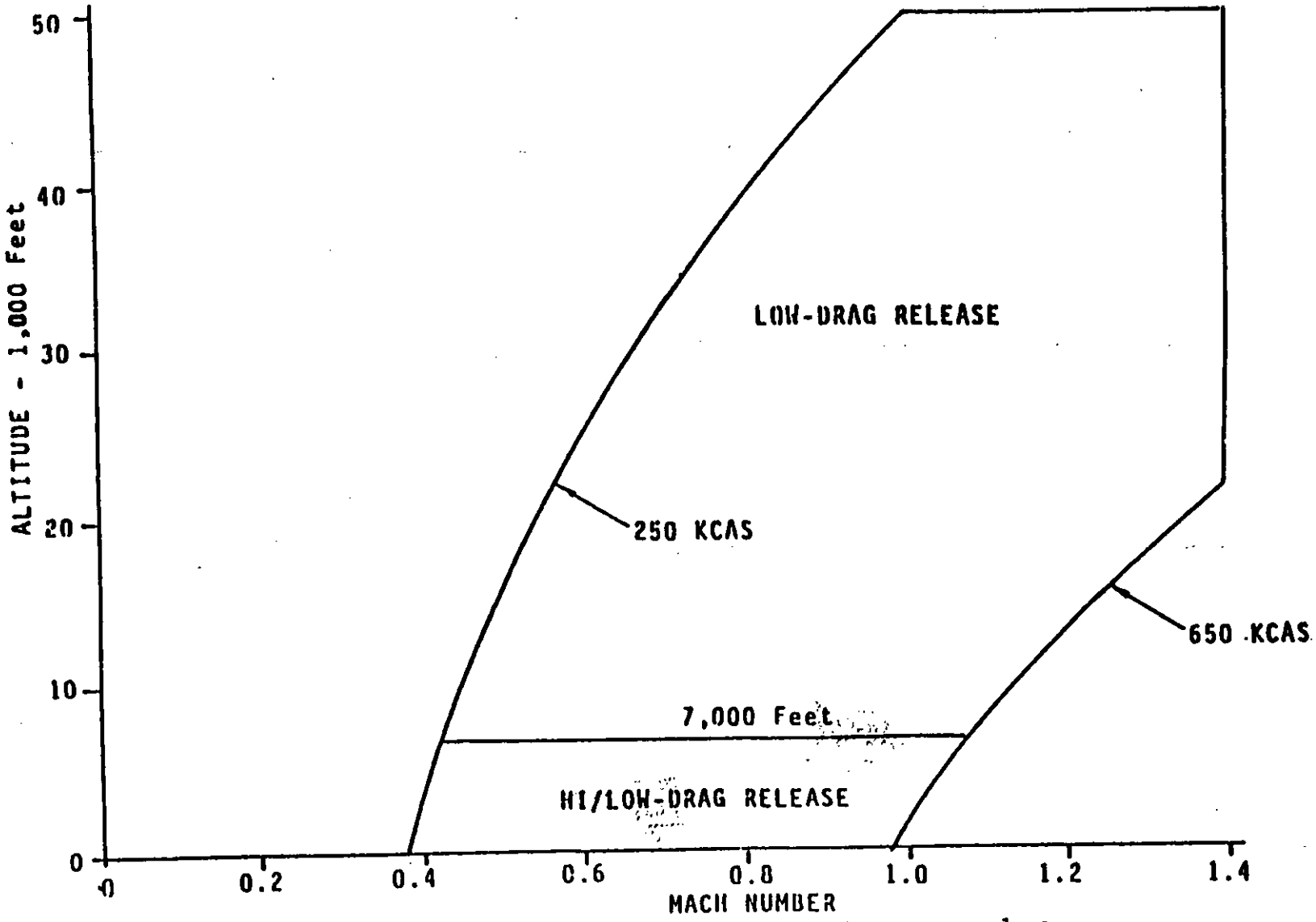


FIGURE 1 - Air carriage/release envelope

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

load. The factor of safety between limit and ultimate load shall be not less than 1.25. All fittings shall include an additional safety factor of 1.2. The factor of safety for castings shall be not less than 1.33. The margin of safety shall be positive for all other structures, seams, and joints. Allowable material stress values shall be as specified in MIL-HDBK-5, MIL-HDBK-17, and MIL-HDBK-23. The item shall have the structural integrity to withstand, without yield, the most severe combined loading condition resulting from the aerodynamic carriage loads specified in FIGURE 1 and the inertial loads specified in MIL-A-8591. The item shall have the structural integrity to withstand, without deformation detrimental to the item performance, the loading condition resulting from the deployment of the item in the high-drag mode. The fins shall be capable of withstanding impact without structual damage attributed to a hung bomb on the centerline stations of the MER and TER bomb rack.

3.2.2 Physical characteristics.

3.2.2.1 Weight and center of gravity. The item weight shall be not greater than 100 pounds. The system center of gravity shall be seven inches plus or minus three inches aft of the forward lug prior to deployment.

3.2.2.2 Dimensions. Dimensions shall be not greater than the 39.30 x 14.57 x 14.57 inches (L x W x H) envelope geometry.

3.2.3 Reliability. The item shall be capable of meeting the performance requirements of 3.2.1.4 with a reliability of 95 percent at a 90 percent confidence level.

3.2.4. Maintainability. The item shall require no scheduled maintenance during long term storage (less than ten years).

3.2.5 Environmental conditions. The item shall be capable of meeting the performance requirements of 3.2.1 during exposure to the pressure-altitude envelope specified in FIGURE 1 and over the temperature range of minus 65 degrees F to plus 160 degrees F. The item shall be capable of meeting the performance requirements of 3.2.1 after exposure to the environments specified in TABLE II.

3.2.6 Transportability. No special handling or transporting devices shall be required.

3.2.7 Shelf life. The item shall have a shelf life goal of ten years when stored using the designed protective packaging (see 6.3.1).

3.2.8 Service life. The item shall have a service life of no less than one year. (See 6.3.2)

3.3 Design and construction.

3.3.1 Material, processes and parts.

3.3.1.1 Nutrient material. Materials that are nutrients for fungi shall not be used where avoidance is possible. When such materials are used and not sealed hermetically, they shall be treated with a fungicidal agent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

TABLE II. Environmental requirements

CONDITIONS	TEST METHOD OR TEST	REMARKS
Altitude	MIL-STD-810, Method 500.1 and 504.1 Procedure 1	Lowest pressure 50,000 ft. above sea level
High Temp	MIL-STD-810, Method 501.1 Procedures I & II	Highest operating temperature 160°F.
Temp Shock	MIL-STD-810, Method 503.1 Procedure I	
Rain	MIL-STD-810, Method 506.1 Procedure I	Wind directions both directions 90° to bomb axis and both directions parallel to bomb axis.
Fungus	MIL-STD-810, Method 508.1 Procedure I	
Salt Fog	MIL-STD-810, Method 509.1 Procedure I	
Dust	MIL-STD-810, Method 510.1 Procedure I	
Acceleration	MIL-STD-810, Method 513.2 Procedure I	
Vibration	MIL-STD-810, Method 514.2 Procedure IIB	
Shock	MIL-STD-810, Method 516.2 Procedure I (FIG. 516.2-1)	
Humidity	MIL-STD-810, Method 507.1 Procedure V	
Catapult & Arrested Landing	MIL-STD-1763 Test 260	Government to conduct tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

3.3.1.2 Standard parts. Standard parts shall be selected in accordance with MIL-STD-965.

3.3.1.3 Corrosion. Materials used shall be corrosion resistant or suitably treated to resist corrosion resulting from environmental conditions encountered during storage and service. The use of dissimilar metals, as specified in MIL-STD-889, in intimate contact with each other shall be avoided unless suitably protected against electrolytic corrosion.

3.3.1.4 Synthetic rubber parts. The use of synthetic rubber parts shall be in accordance with MIL-STD-1523.

3.3.2 Identification plates and product marking. Identification plates and product marking shall be in accordance with MIL-STD-130.

3.3.3 Interchangeability. Like components of the item shall be physically and functionally interchangeable as specified in MIL-I-8500.

3.3.4 Safety. The item shall be capable of meeting the safety requirements specified below:

a. The item shall not degrade or negate the safety features of the MK 83 through the flight envelope of FIGURE 1.

b. The item shall not degrade or negate the safety features of the MK 83 throughout the maintenance, storage, shipping, and handling environments.

c. The item shall not prevent the MK 83 from being rendered safe, unless specifically exempted for operational purposes.

d. The item shall not prevent the MK 83 from being jettisoned in flight in an unarmed condition upon demand.

e. Flight data shall be taken to determine the safe operating envelope under the failure mode condition of a hung centerline store on the MER and TER bomb racks.

f. The retarder shall demonstrate a 1×10^{-6} probability of not separating from the fin canister over its operational envelope.

g. The probability of the occurrence of a hazard severity category I condition, as specified in MIL-STD-882, during any captive flight of one hour shall be not greater than 1×10^{-6} per item. The probability of getting a low drag configuration when high drag is selected shall be no greater than 1×10^{-6} per single release. The hazard probability of an inadvertent deployment of the retarder during any captive flight of one hour shall be not greater than 1×10^{-6} per item. The method for determining the probability of inadvertant deployment shall be documented for review by the procuring activity. (see 6.2.2)

3.3.5 Human performance/human engineering. The item shall be capable of meeting the requirements of MIL-H-46855 and MIL-STD-1472.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

3.4 Documentation.

3.4.1 Design approval drawings. Design approval drawings shall be prepared and made available to the procuring activity. (see 6.2.2)

3.4.2 Test equipment calibration. Test equipment shall be calibrated and maintained in accordance with MIL-STD-45662. When specified in the contract or purchase order (see 6.2.1), the contractor shall prepare a calibration plan for procuring activity review. (see 6.2.2)

3.4.3 Test plan and test procedures. When specified in the contract or purchase order (see 6.2.1), the contractor shall prepare a test plan and test procedures for procuring activity review. (see 6.2.2) The test plan and procedures shall include verification of structural adequacy.

3.5 Logistics.

3.5.1 Maintenance. The item shall require no scheduled maintenance for the shelf life specified in 3.2.7 and the service life specified in 3.2.8. The item shall require no corrective/unscheduled maintenance other than visual inspection, external corrosion control, and external rehabilitation such as touch-up painting, remarking, and repair or replacement of the lanyard assembly. Corrective/unscheduled maintenance shall require only ordinary hand tools.

3.5.2 Supply. The item shall be capable of routine entry into the federal supply system.

3.5.3 Facilities and facility equipment. The item shall require no special facilities or ancillary equipment to support the item in the field.

3.5.4 Special tools. The item shall require no special tools for attaching to the MK 83 bomb body.

3.6 Personnel and training. The design and use of the item shall be such that no special ordnance or maintenance skills are required. Training shall be accomplished as on-the-job training with reference to technical order information.

3.7 Major component characteristics. Not applicable.

3.8 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.2.3).

3.9 Workmanship. The item shall be constructed and finished with particular attention given to neatness and thoroughness of marking of parts, welding and brazing, plating, riveting, finishes, and machine-screw assemblage. The item shall be free of burrs, sharp edges, dirt, chips, or other foreign matter and visible defects which are likely to affect or render the item unsatisfactory and nonserviceable over the useful life for the operation or function intended. Welds shall be in accordance with MIL-W-8611.

4. QUALITY ASSURANCE PROVISIONS.

4.1. Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special tests and examinations.

4.2.1 Selection of parts, components, or equipments. When specified in the contract or purchase order (see 6.2.1), the contractor shall conduct special tests or examinations for qualification evaluation for selection of parts, components, or equipments.

4.2.2 Static load tests. The item shall be subjected to static load tests to verify its capability of meeting the structural adequacy requirements. The static load tests shall be conducted on but not limited to; the attachment mechanism for affixing the item to the MK 83 bomb, the stabilizer structure, and the attachment mechanism for affixing the high-drag retarder to the stabilizer structure.

4.2.3 Environmental tests. The item shall be subjected to the environmental tests specified in TABLE II to simulate the environmental stresses that occur. Any one item shall be subjected to single environmental tests or one of the following sequential tests.

Sequence A: Temperature and humidity, shock, vibration

Sequence B: Salt fog, dust, rain, acceleration, high temperature

Sequence C: Temperature and humidity, shock, fungus

Sequence D: Altitude, temperature shock, vibration

Following each test or test sequence, the item shall be subjected to a static function test to verify proper function of latches, and cover release, and to demonstrate that the forces required for deployment of the retarder are not greater than the forces available during the lowest required airspeed.

4.2.4 Low-drag accuracy flight tests.

4.2.4.1 Flight tests. The system shall be flight tested in the low-drag mode to verify the ballistic dispersion, and flight safety requirements.

4.2.4.2 Procedure to determine low-drag CEP. The normal plane CEP shall be determined in the plane perpendicular to the computed bomb trajectory and passing through the computed mean impact. The computed mean impact shall be determined by starting the computed trajectory on the bomb two seconds after release. The miss distance shall be converted to mils CEP in the plane perpendicular to the fight path at impact before computing the standard deviations. The low-drag CEP shall be based on system performance utilizing the standard (non-thermally protected) MK 83 bomb bodies, when released within the 300 to 650 KCAS speed regime. The following equation shall be used to determine CEP:

Source: https://assist.dla.mil - Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

$$CEP = 1.1774\sigma$$

where σ is the standard deviation of the radial miss distances,

$$\text{that is: } \sigma = \left[\frac{\sum_{i}^{n}(Xi - \bar{X})^2}{n - 1}\right]^{1/2} \quad \text{where} \quad \begin{array}{l} n = \text{total number of test samples} \\ Xi = \text{radial miss distance} \\ \bar{X} = \text{mean radial miss distance} \end{array}$$

4.2.4.3 Censoring. Four percent of the total number of drops may be removed from the test sample before the computation of CEP, after which no further censoring shall be permitted.

4.2.5 High drag accuracy flight tests.

4.2.5.1 Flight tests. The system shall be flight tested in the high-drag mode to verify the ballistic dispersion and flight safety requirements.

4.2.5.2 Procedure to determine high-drag CEP. The normal plane CEP shall be determined in the plane perpendicular to the computed bomb trajectory and passing through the computed mean impact. The computed mean impact shall be determined from the release point on the launch aircraft to the position of the bomb after no more than 1000 feet of vertical fall. The miss distance shall be converted to mils CEP in the plane perpendicular to the flight path at impact before computing the standard deviations. The high-drag CEP shall be based on system performance when released within the 300 to 650 KCAS speed regime. The following equation shall be used to determine CEP:

$$CEP = 1.1774\sigma$$

where σ is the standard deviation of range errors and deflection errors, respectively.

$$\text{that is: } \sigma = \left[\frac{\sum_{i}^{n}(Xi - \bar{X})^2}{n - 1}\right]^{1/2} \quad \text{where} \quad \begin{array}{l} n = \text{total number of test samples} \\ Xi = \text{radial miss distance} \\ \bar{X} = \text{mean radial miss distance} \end{array}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

4.2.5.3 Censoring. Four percent of the total number of drops may be removed from the test sample before the computation of CEP, after which no further censoring shall be permitted. Aircraft rack malfunctions which are classified as "No Tests" are excluded from the test sample prior to the four percent censoring.

4.2.6 Demonstrations and fit tests. During accomplishment of the following tests, handling and transportation equipment used for MK-83 GP bombs shall be used to demonstrate compatibility. Fit tests shall be accomplished in accordance with MIL-STD-1289.

4.2.6.1 Test plan. The item and system shall undergo a planned sequence of tests to demonstrate interfaces and functions with the bomb racks and aircraft listed in TABLE I.

4.2.6.2 Location of tests. Non-flight tests shall be conducted at a facility acceptable to the procuring activity or at a Government facility specified in the contract. (see 6.2.1)

4.2.6.3 Test conditions. In tests applicable to referenced military standards or other referenced documents, the test conditions specified therein shall apply unless modified herein. When tests not applicable to referenced military standards or other documents are required, the following general environmental conditions shall apply:

a. Temperature: 55 to 99°F

b. Humidity: 0 to 95 percent, relative

c. Altitude: At test site

d. Vibration: None

4.2.6.4 Interface tests. The item shall be assembled to a MK 83 bomb to verify the attaching ring to bomb interface.

4.2.6.4.1 Launch aircraft and rack interface. The system shall be compatible with the bomb rack types and aircraft listed in TABLE I. Fit tests shall be accomplished with the racks and aircraft specified in the contract or purchase order. (see 6.2.1)

4.2.6.4.2 Accessory interface. Fuzes and arming wire installations, settings, operations, access, and removal shall be demonstrated.

4.2.6.4.3 Rack ejection tests. The system shall be subjected to aircraft rack ejection tests to verify the lanyard system interface and function requirements. Rack ejection tests shall be conducted on representative bomb rack types listed in TABLE I. The system shall be assembled and the lanyard rigged for retarder deployment. The lanyard attachment link at the arming solenoid shall be instrumented to record lanyard pull force. A special centerline hung store failure mode test shall be conducted to determine fin impact resistance and bomb and lanyard response under these conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

4.2.6.5 Transportability Verification. The transportability of the item shall be verified by inspection of referenced drawings, hardware, and demonstration.

4.2.7 Flight tests.

4.2.7.1 Test aircraft and facilities. The system shall be flight tested using the aircraft and bomb rack types listed in TABLE I. The tests shall be conducted from aircraft and at test facilities designated by the government. (see 6.2.1)

4.2.8 Safety. The system shall be evaluated for safety requirements during the development and testing programs. Both high and low drag operational characteristics of the system shall be evaluated. The requirements of 3.3.4 shall be verified by analysis and data from flight tests.

4.2.9 Reliability.

4.2.9.1 Demonstration. The reliability shall be demonstrated at the 90 percent confidence level. Any system released in which the retarder does not deploy and meet the performance requirements of 3.2.1.4 shall be counted as a failure.

4.2.9.2 Shelf Life. Item reliability after simulated ten year storage period in the designed packaging shall be determined by analysis of test data.

4.2.9.3 Service Life. Item reliability after one year of open storage shall be determined by analysis of test data.

4.2.10 Maintainability. The maintainability requirement of 3.2.4 shall be demonstrated.

4.3 Quality conformance inspection. Quality conformance test and inspection criteria defined herein include the tests and inspections specified in TABLE III. The following physical characteristics shall be checked by inspection of referenced drawings or parts:

a. Weight and Center of Gravity (CG)
b. Dimensions
c. Nameplate and Product Markings
d. Workmanship

4.4 First article inspection. The first article inspection shall be conducted during the Low Rate Initial Production (LRIP) and shall assure that the contractor has conformed to all items of the procurement data package. Flight tests shall be performed on 24 of the 7000 units to verify the performance specified in section 3.2.1 (a), (b), (f), and (g).

5. PACKAGING

5.1 Presentation, packaging, and packing. The item shall be preserved, packaged and packed in accordance with FED-STD-102. Level C packaging of the items shall meet the requirements of MIL-STD-794 and MIL-P-9024.

5.2 Marking. In addition to the markings specified by the contract or purchase order, packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

TABLE III. Specification requirements/verification cross reference index

REQUIREMENTS	DESCRIPTION	VERIFICATION	N/A	INSPECTION	METHODS ANALYSIS	DEMONSTRATION	TEST
3.	REQUIREMENTS	-	x				
3.1	Item description	-	x				
3.1.1	Item diagrams	-	x				
3.1.2	Interface description	-	x				
3.1.2.1	Item to MK 83 Bomb	4.2.6.4				x	
3.1.2.2	Item to launch aircraft	4.2.6.4.1, 4.2.6.4.3				x	x
3.1.2.3	Accessory physical interface	4.2.6.4.2				x	
3.1.2.4	Fuze installation	4.2.6.4.2				x	
3.1.2.5	Fuze function	4.2.8					x
3.1.3	Major component list	-	x				
3.1.4	Government furnished property list	-	x				
3.1.5	Government loaned property list	-	x				
3.2	Characteristics	-	x				
3.2.1	Performance	-	x				
3.2.1.1	Flight	4.2.4, 4.2.5, 4.2.8					x
3.2.1.2	Temperature	4.2.3, 4.2.8					x
3.2.1.3	Aircraft separation	4.2.4, 4.2.5 4.2.8					x
3.2.1.4	Retarder deployment	4.2.5, 4.2.8					x
3.2.1.5	Accuracy	4.2.4, 4.2.5			x		x

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

TABLE III. Specification requirements/verification cross reference index (Continued)

REQUIREMENTS	DESCRIPTION	VERIFICATION	N/A	INSPECTION	METHODS ANALYSIS	DEMONSTRATION	TEST
3.2.1.6	Structural adequacy	4.2.2			x		x
3.2.2	Physical characteristics	-	x				
3.2.2.1	Weight and center of gravity	4.3			x		x
3.2.2.2	Dimensions	4.3		x		x	
3.2.3	Reliability	4.2.9			x		x
3.2.4	Maintainability	4.2.10			x		
3.2.5	Environmental conditions (Table II)	4.2.3			x	x	x
3.2.6	Transportability	4.2.6.5		x	x	x	
3.2.7	Shelf life	4.2.9.2			x		
3.2.8	Service life	4.2.9.3			x		
3.3	Design and construction	-	x				
3.3.1	Materials, processes and parts	-	x				
3.3.1.1	Nutrient material	4.2.1			x		
3.3.1.2	Standard parts	4.2.1		x			
3.3.1.3	Corrosion	4.2.1		x			
3.3.1.4	Synthetic rubber parts	4.2.1		x			
3.3.2	Electromagnetic radiation	-	x				
3.3.3	Identification plates & product marking	4.3		x			
3.3.4	Interchangeability	4.2.1		x			
3.3.5	Safety	4.2.8		x	x	x	x
3.3.6	Human performance/human engr'g	-	x				
3.4	Documentation	4.3		x			

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

TABLE III. Specification requirements/verification cross reference index (Continued)

REQUIREMENTS	DESCRIPTION	VERIFICATION	N/A	INSPECTION	METHODS ANALYSIS	DEMONSTRATION	TEST
3.5	Logistics	-	x				
3.5.1	Maintenance	4.2.10			x		
3.5.2	Supply	-	x				
3.5.3	Facilities & facility equipment	-	x				
3.5.4	Special tools	4.2.6			x		
3.6	Personnel and Training	-	x			x	
3.7	Major component characteristics	-	x				
3.8	First Article	4.4, 6.2.3		x		x	x
3.9	Workmanship	4.3		x			

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. This item is intended for use with a free-fall MK 83 general purpose bomb which is capable of supersonic carriage and release in either a high or low drag configuration from tactical or strategic aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Number, title and date of this specification.
b. Responsibility for inspection if other than as specified in 4.1.
c. Calibration plan, if required (see 3.4.2)
d. Test plan and test procedures, if required (see 3.4.3)
e. Special tests for parts, components, equipment selection, if required (see 4.2.1).
f. Location of Government facility for nonfight tests, if required (see 4.2.6.2).
g. Racks and aircraft for fit tests (see 4.2.6.4.1).
h. Test aircraft and Government facility for flight tests (see 4.2.7.1)
i. Level of packaging (see 5.1)
j. Marking (see 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID No.	Option
3.3.5	Systems Safety Program Plan	DI-H-7047	-
3.4.1	Design Approval Drawings	DI-E-7031	-
3.4.2	Calibration Plan	DI-R-7065	-
3.4.3	Test Plans/Procedures	DI-T-5204	-

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSOL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. When a first article inspection is required, the item will be tested and should be a sample selected from the first production items. The first article should consist of 24 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

6.3 Definitions.

6.3.1 Shelf Life. The period of time beginning with the date of manufacture that an item is capable of remaining in the combined wholesale (including manufacturer) and retail storage system and still remain suitable for issue to the user.

6.3.2 Service Life. The length of time the item is capable of remaining outside its storage container in unimproved open storage in all climatic zones without its reliability being degraded below the requirements of this specification.

Preparing Activity
Navy - AS
(Project 1325-N110)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-85596(AS)	Fin, Retarder, Air Inflatable, BSU-85/B

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:11Z
Check the source to verify that this is the current version before use.